IIT- Tutorial Lab Week 11 – Testing & Exception Handling

Introduction

In this week's tutorial, we will start Testing and Exception Handling. This is important preparation work for your Assignment number 4.

Part 1: Exception Handling

a) Each of the following statements, generates one of the Traceback error messages (labelled (a) through (t) in Table 1. Determine the error message generated by each statement.

1. `x = str(asdf)` **f)**
2. `f = open("abc.txt", 'R')` **k)**
3. `str = abs("str")` **L)**
4. `total = ('2' * '3')` **j)**
5. `x = ['a', 'b', 'c'][]` **d)**
6. `x = list(range(1, 9, '1'))[8]` **e)**
7. `x = '23'`
   `print(x.startswith(2))` **i)**
8. `x = '8'`
   `x.append(2)` **c)**
9. `{'1':"uno", 2:"dos"}['2']` **s)**
10. `{"Mars":"War","Neptune":"Sea"}.values()[2]` **b)**
11. `num = [1, 3].remove(2)` **0)**
12. `num = ('1', '3').index(3)` **a)**
13. `letter = ("ha" * '5')[9]` **j)**
14. `s = ['s', 'e', 'd']['0']` **g)**
15. `x = {1, 2, 3}[1]` **n)**
16. `(2, 3, 1).insert(0)` **t)**
17. `num = eval('x = 3*3')` **d)**
18. `value = min(1, 'a')[1]` **q**
19. `del ['11', '12', '13'][0][0]` **h)**
20. `print([2] in {1: [2], 2: [3], 3: [1]})` **m)**
21. `["air", "fire", "earth", "water"].sort()[2]` **r)**
22. `"1, 2, 3".find(1)` **p)**

```
(a) ValueError: tuple.index(x): x not in tuple
(b) TypeError: 'dict_values' object does not support indexing
(c) AttributeError: 'str' object has no attribute 'append'
(d) SyntaxError: invalid syntax
(e) TypeError: 'str' object cannot be interpreted as an integer
(f) NameError: name 'asdf' is not defined
(g) TypeError: list indices must be integers, not str
(h) TypeError: 'str' object doesn't support item deletion
(i) TypeError: startswith first arg must be str or a tuple of str, not int
(j) TypeError: can't multiply sequence by non-int of type 'str'
(k) ValueError: invalid mode: 'R'
(l) TypeError: bad operand type for abs(): 'str'
(m) TypeError: unhashable type: 'list'
(n) TypeError: 'set' object does not support indexing
(o) ValueError: list.remove(x): x not in list
(p) TypeError: Can't convert 'int' object to str implicitly
(q) TypeError: unorderable types: str() < int()
(r) TypeError: 'NoneType' object is not subscriptable
(s) KeyError: '2'
(t) AttributeError: 'tuple' object has no attribute 'insert'
```

b) In the following exercises, determine the output displayed by the lines of code.

23.
```
# Assume the user enters a letter.
try:
    num = float(input("Enter a number: "))
    print("Your number is", num)
except:
    print("You must enter a number.")
```

Enter a number: a
You must enter a number

24.
```
nafta = ["Canada", "United States", "Mexico"]
try:
    print("The third member of NAFTA is", nafta[3])
except IndexError:
    print("Error occurred.")
```

Error occurred

25.
```
flower = "Bougainvillea"
try:
    lastLetter = flower[13]
    print(lastLetter)
except TypeError:
    print("Error occurred.")
except IndexError as exc:
    print(exc)
    print("Oops")
```

String index out of range
Oops

26. Assume that the file **Ages.txt** is located in the current folder and the first line of the file is **Twenty-one\n**.

```
try:
    infile = open("Ages.txt", 'r')  # FileNotFound if fails
    age = int(infile.readline())    # ValueError if fails
    print("Age:", age)
except FileNotFoundError:
    print("File Ages.txt not found.")
except ValueError:
    print("File Ages.txt contains an invalid age.")
    infile.close()
else:
    infile.close()
```

File Ages.txt contains an invalid age

27. Assume that the file **Salaries.txt** is located in the current folder and the first line of the file contains the string 20,000.

```
def main():
    try:
        infile = open("Salaries.txt", 'r')  # FileNotFound if fails
        salary = int(infile.readline())     # ValueError if fails
        print("Salary:", salary)
    except FileNotFoundError:
        print("File Salaries.txt not found.")
    except ValueError:
        print("File Salaries.txt contains an invalid salary.")
        infile.close()
    else:
        infile.close()


    finally:
        print("Thank you for using our program.")

main()
```

File Salaries.txt contains an invalid salary
Thank you for using our program

File Ages.txt not found

28. Redo Exercise 26 with the assumption that the file **Ages.txt** is not located in the same folder as the program.

29. The following program will perform properly if the user enters 0 in response to the request for input. However, the program will crash if the user responds with "eight". Rewrite the program using a **try/except** statement so that it will handle both types of responses. See Fig. 6.1.

```
while True:
    try:
    n = int(input("Enter a nonzero integer: "))
    if n != 0:
        reciprocal = 1 / n
        print("The reciprocal of {0} is {1:,.3f}".format(n, reciprocal))
        break
    else:
        print("You entered zero. Try again.")
```

print("You entered zero. Try again.")

except valueError:
print("You did not enter a nonzero integer. Try again.")

```
Enter a nonzero integer: 0
You entered zero. Try again.
Enter a nonzero integer: eight
You did not enter a nonzero integer. Try again.
Enter a nonzero integer: 8
The reciprocal of 8 is 0.125
```

Part 2: Testing

For each of the following business rules (see page 5), add test data to the table to thoroughly test the program. **Give each test a unique description**. No two tests should have the same description and **avoid numbers in your descriptions**. Note that the number of rows in the table below is no indication on how many tests you should have. You may need more, you may need fewer. You must first work out the equivalence partitions, then type of data the program works with (ordinal vs. non-ordinal), then the boundaries and special cases, before you start filling in the table with the actual tests.

a) Movie tickets cost $5 for a child (<10 years of age), $10 for a teenager (10yr ≤ age < 20yr) and $12 for an adult (≥ 20yr).

Description	Age	Ticket cost()
Young child	9	5
Teenager at lower boundary	10	10
Teenager just before adult boundary	19	10
Adult at lower boundary	20	12

b) A company has a recruiting policy of only interviewing applicants between 25 and 40 inclusive. However, they will interview applicants up to the age of 45 if they have at least 10 years' experience.

Description	Age	Experience	interview()
Below minimum age	24	5	No
At minimum age	25	9	Yes
At maximum standard age	40	0	Yes
Above standard age with enough experience	41	9	No
At maximum extended age with required experience	45	10	Yes
Above maximum extended age	46	11	No

c) To be elected, a candidate must get over half of the votes cast. An election has 3 candidates, A, B, and C. If no candidate gets over half of the votes, the result is "-". The input here is the actual number of votes, not percentages.

Description	votes A	votes B	votes C	result()
Candidate A wins	100	0	0	A
Candidate B wins	0	100	0	B
Candidate C wins	0	0	100	C
A candidate votes higher than B&C	50	25	24	A
B&C is higher than A candidate	50	25	26	-